Golo AI Assistant — Features & Specification Catalog

Technical Inventory of Active Production Features & Planned Roadmap Objectives for the Merchant Web Platform

1. FRONTEND SPECIFICATIONS & FEATURE LIST (Currently Implemented & Active)

Feature Name	Technical Specification & User Experience	Status
Search Bar (Query Input)	Full-width text input with 2,000 char capacity, dynamic warning counter at 1800+ chars, auto-grow, focus ring, send on Enter key or dispatch button.	LIVE
Dark & Light Mode	Default Dark Mode (#0f172a/#1e293b). Dedicated Theme Preference dropdown (Default Dark / Default Light) with localStorage persistence & 1-click Sun/Moon toggle.	LIVE
Window Controls (Min / Max)	Three dedicated controls: **Minimize [-]** collapses to bottom-right floating dock; **Maximize / Restore [Expand]** opens 940x820 dual-pane workspace; **Close [X]** dismisses.	LIVE
3D WebGL Models & Visuals	Interactive 3D WebGL particle constellation with dual orbital rings reacting to cursor tilt in hero section; live rotating 3D mini-orb in chat header indicating AI reasoning.	LIVE
Floating Launcher Button	64x64px circular trigger in brand royal blue (#3B7DDD) with pulsing beacon, hover tooltip ("Ask Golo AI"), and green online status dot.	LIVE
Suggestion Chips	One-click pill buttons below welcome prompt for common merchant questions (Add product, Payout schedule, KYC verification, Shipping label). Dispatches instantly on click.	LIVE
App Module Navigator	Integrated sidebar in Maximized Pro view providing direct shortcuts to Products, Payments, KYC, Orders, Promotions, and Analytics modules.	LIVE
Response Formatter & Copy	Numbered step-by-step guidance, bold UI paths (e.g. **Dashboard > Products**), code tags, source attribution badge, and 1-click clipboard copy button on hover.	LIVE
Typing State Indicator	Three-dot bouncing animation indicating sub-100ms processing delay, with input field lock to prevent accidental duplicate queries.	LIVE
Support Escalation Card	Specialized UI card (State 7) auto-generating Support Ticket #REF-XXXX, estimated 4-hour SLA follow-up, and ticket status button.	LIVE
CSAT 5-Star Rating Widget	Interactive 5-star rating widget (State 8) collecting user feedback with submit/skip options, linked to session and merchant ID.	LIVE
Zero Sign-In Barrier	Frictionless, open merchant access. No sign-in or sign-out prompts required to interact with the AI assistant.	LIVE

2. BACKEND SPECIFICATIONS & ARCHITECTURE (Currently Implemented & Active)

Core Architecture	Technical Specification & Algorithmic Design	Status
Dual-Path Routing	Path A: Exact FAQ match via tokenized similarity (<15ms, $0 cost). Path B: Semantic context search over 40+ app modules for complex queries.	LIVE
Grounded Knowledge Base	65+ pre-loaded verified FAQs across 10 operational domains: Product Listings, Payments, KYC, Orders, Policies, Promotions, Technical, Settings, Analytics, Onboarding.	LIVE
Strict Out-of-Scope Filter	Strict domain filter rejecting unrelated queries (weather, recipes, general trivia) with exact message: *"It is not our app related question. Please try again."*	LIVE
Security & Sanitization	Regex pattern filters blocking prompt injections (jailbreak, role-play overrides), PII redaction for passwords/OTPs/cards, and offensive language filtering.	LIVE
Per-Merchant Rate Limiter	Token-bucket limiter enforcing 20 RPM and 200 RPD quotas per merchant_id with countdown notification messages upon reaching limits.	LIVE
Multi-Turn Session Memory	Sliding window preserving last 10 conversation turns per session_id for natural multi-turn contextual continuity.	LIVE
REST API Gateway	FastAPI server running with 7 endpoints: POST /chat, GET /health, GET /welcome, GET /faqs, POST /ratings, POST /escalate, GET /merchant/{id}/usage.	LIVE
Dual-Mode Redundancy	Live REST API execution with automatic client-side embedded JavaScript AI engine fallback if the server is offline.	LIVE

3. FUNCTIONALITIES ACCEPTING MERCHANT ID AS INPUT (Specific Architecture)

The system specifically accepts and processes **merchant_id** (e.g., "MRC-8821") across 5 core workflows to ensure customized, quota-controlled, and auditable merchant support:

Functionality / API Route	Input Signature	Specific Role & Processing Logic of merchant_id
1. Conversational Chat POST /chat	{query, session_id, **merchant_id**}	Tags chat telemetry to merchant account; enables store-specific context injection; echoes merchant_id in response metadata for auditability.
2. Token-Bucket Rate Limiter ai_engine.process_query	merchant_id: str	Enforces strict tier quotas: **20 RPM** (Requests Per Minute) sliding window and **200 RPD** (Requests Per Day). Prevents API exhaustion.
3. Live Quota Telemetry GET /merchant/{id}/usage	merchant_id (URL Path)	Returns real-time consumption metrics: rpm_used, rpm_remaining, rpd_used, rpd_remaining, and lifetime query history.
4. Priority Support Escalation POST /escalate	{session_id, reason, details, **merchant_id**}	Binds auto-generated Support Ticket (e.g. **#REF-5551**) to verified merchant_id; packages last 4 conversational turns for 4-hour human SLA.
5. CSAT Satisfaction Survey POST /ratings	{session_id, rating, comment, **merchant_id**}	Logs 1-to-5 star satisfaction feedback against merchant account (hashed as merchant_id_hash for privacy) to evaluate seller happiness trends.

4. FUTURE ROADMAP & PLANNED SPECIFICATIONS (Objectives for Later Implementation)

The following advanced features are specified in the system design and scheduled for subsequent modular development phases:

Feature Name	Technical Specification & Scope	Target Phase
• Mic Option (Voice Input)	Integrated Web Speech API / Whisper transcription button beside input bar allowing hands-free voice query dictation in real-time.	Phase 7 Objective
• In-Chat Search Bar	Keyword search bar at the top of the message stream enabling merchants to search historical conversation turns and jump directly to past answers.	Phase 7 Objective
• Chat History Drawer	Collapsible multi-session archive drawer allowing merchants to switch between past support conversations, export transcripts, or clear history.	Phase 7 Objective
• Merchant Profile Section	In-widget profile modal displaying active Store Name, Merchant Tier (Gold/Standard), KYC Status badge, and direct links to Store Settings.	Phase 8 Objective
• Multilingual Voice (TTS)	Text-to-speech voice playback for generated instructions in English, Hindi, Tamil, Telugu, and Bengali for regional merchant accessibility.	Phase 8 Objective
• OCR Document Scanner	In-chat drag-and-drop file attachment enabling merchants to upload GST certificates, shipping slips, or error screenshots for automatic parsing.	Phase 8 Objective
• Proactive Stock Alerts	Real-time push notifications inside the chat widget alerting merchants when product stock drops below threshold or payouts are disbursed.	Phase 9 Objective
• Live Helpdesk Webhooks	Bi-directional webhooks integrating ticket escalations directly into Zendesk, Freshdesk, and Jira Service Management with live status sync.	Phase 9 Objective

Summary Specification Note: *Current production version v1.0.0 delivers 100% of core grounded guidance, 3D WebGL visuals, dual-path FAQ matching, window controls, and rate limiting with zero code disruption. Future roadmap objectives integrate modularly into the existing architecture.*